

Does a Slime Mold Think?

What counts as cognition — and does biology matter?

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Computational Modelling for Psychology & Neuroscience

Where we've been

- ▶ **Dog · bell · food** — nobody argues. We grant “it learned.”
- ▶ **Sunflower · sun · sugar** — anticipation of dawn, but *built in*. Not learned.
- ▶ **Pea plant · fan · light** — *claimed* associative learning. Contested; a replication failed.

Notice what moved as we went down the list: not the evidence so much as **how much evidence we demanded**. Our credence tracked the *material*, not the data.

Ask: If the same conditioning result showed up in a rat, would you have asked for as much proof?

Now a case that clears the behavioural bar cleanly — and has **no neurons at all**.

A single cell with no brain

Physarum polycephalum — a slime mould.

- ▶ One giant **multinucleate cell**. No neurons, no synapses, no nervous system.
- ▶ Forages by growing a living network of tubes; routes are reinforced or pruned by internal **flow feedback**.
- ▶ Long filed under “fungus”; actually an amoeba.

Everything on the next slides is done by *this*.

It solves mazes

Place it between two food sources in a maze. It fills the maze, then **withdraws from dead ends** and keeps the tube along the **shortest path** connecting the food.

Ask: That's problem-solving. Is it cognition — or just the physics of a flow network settling into a minimum?

— *Nakagaki, Yamada & Tóth (2000), Nature*

It designs networks that rival engineers

Arrange oat flakes as the cities of Greater Tokyo. The mould grows a network that **approximates the real rail system** — balancing total tube length, transport efficiency, and **fault tolerance** against accidental breaks.

Those are the exact tradeoffs human planners spend months negotiating.

Ask: We assume optimization implies a designer with a goal. Does this?

— *Tero et al. (2010), Science*

But is it *learning*?

The sunflower failed this test. The slime mould does not.

- ▶ **Habituation** — made to cross a bridge laced with a *harmless* bitter substance, it hesitates at first, then stops reacting over days — and recovers if left alone. Habituation is a textbook elementary form of learning.
- ▶ **Anticipation** — shocked with cold at fixed intervals, it slows *before* the next shock — and still slows once at the expected time when the shock is withheld.

Ask: “Learning needs a nervous system.” Here it is without one.
So is the rule wrong, or was the behaviour never learning?

— *Boisseau, Vogel & Dussutour (2016); Saigusa et al. (2008)*

Memory — outside the body

It avoids ground it has already covered by **sensing the extracellular slime it left behind**. The record of where it has been is written into the *environment*, not stored in a head.

Ask: We locate cognition “in the brain.” What do we do with a memory that lives outside the organism?

— *Reid, Latty, Dussutour & Beaton (2012), PNAS*

So: which definition are you using?

No neurons. No brain. One cell. Yet: maze-solving, network optimization, habituation, anticipation, externalized memory.

Two ways to draw the line — and they disagree:

- ▶ **Functional** — cognition is what a system *does* (solve, learn, remember). → It qualifies.
- ▶ **Substrate** — cognition is a property of certain *stuff* / *architecture* (nervous systems). → It doesn't.

Same axis as the pea plant, one notch harder — because now the behaviour really is learning. **There is no neutral seat.** Pick one and pay for it.

The twist for *this* course

Tero's team didn't just watch the mould. They abstracted its behaviour into a **mathematical model** — reinforce tubes carrying more flow, starve the rest — now used to design real-world networks. Others proved formally that *Physarum* **computes shortest paths**.

So the organism *is* an algorithm.

“Does the slime mould think?” collapses into “**is the computation it runs cognition?**” — which is the question this entire course exists to pick apart.

— *Tero et al. (2010); Bonifaci, Mehlhorn & Varma (2012)*

The whole arc on four axes

“X learns that Y predicts Z” hides four separate claims. Line the cases up against them:

Does the case have...	Dog	Sunflower	Pea	Slime mould	LLM
a real $Y \rightarrow Z$ contingency	yes	yes	yes	yes	yes
behaviour that tracks it	yes	yes	<i>claimed</i>	yes	yes
acquisition by experience	yes	no (built-in)	<i>claimed</i>	yes	train: yes / run: no
an internal representation	yes?	a clock?	<i>claimed</i>	external?	<i>contested</i>

- ▶ The **top two rows are free** — a thermostat passes them. “Responds to a predictor” does no work.
- ▶ **Acquisition** is where *learning* actually lives: it splits the sunflower (no) from the dog and slime mould (yes), and cuts the LLM in half — trained, then frozen.
- ▶ **Representation** — the row people *assume* defines cognition — is the one we can least settle from behaviour. Nearly every cell is a question mark.
- ▶ Read the **Pea** and **Dog** columns top to bottom: almost identical. The gap is a failed replication and a hunch about what plants are made of. **That gap is the bias.**

(A fifth axis — is Z a reward (a reinforcer) or merely a benefit? — fractures the same way: food, sugar, light, loss-minimization, an RLHF reward model. Different things, one word.)

Your turn — no roles, no notes

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To argue it, decide what you're actually tracking:

- ▶ What evidence would **change your mind** — in either direction?
- ▶ Does it matter that it's **alive**?
- ▶ If the **identical algorithm** ran in silicon, would your answer change? If yes, your definition is tracking *biology*. If no, it's tracking *computation*. Which did you mean?

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